VERIFIED EXISTING BASELINE

EfficientNet-B0 V3 model evidence

Four-folder fundus image classification · held-out image-level evaluation · not cataract grading

SCOPE Real measured results from the saved epoch-28 checkpoint. Development evidence only; not clinical validation and never used to assign urgency.

Held-out test results

Metric	Final test (n=91)
Accuracy	72.5275%
Balanced accuracy	62.2917%
Macro F1	66.0338%
Weighted F1	70.6214%
Macro OvR ROC-AUC	88.9266%

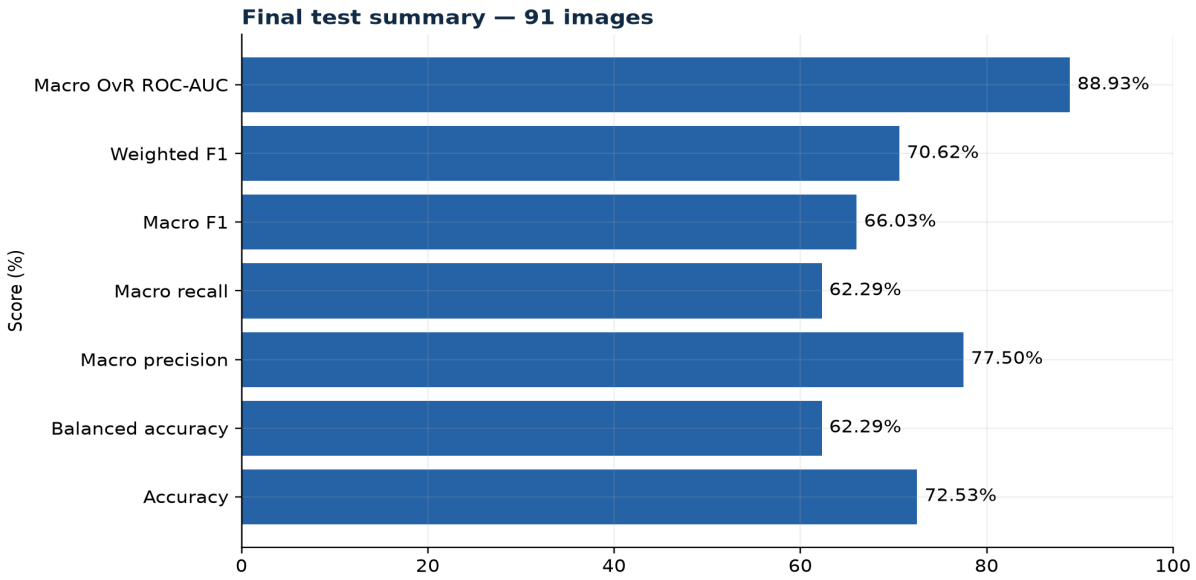


Figure 1. Final test metrics regenerated from saved evaluation artifacts.

EVALUATION FIGURES

Error structure and discrimination

Exact confusion matrix and one-vs-rest ROC curves from saved final-test predictions

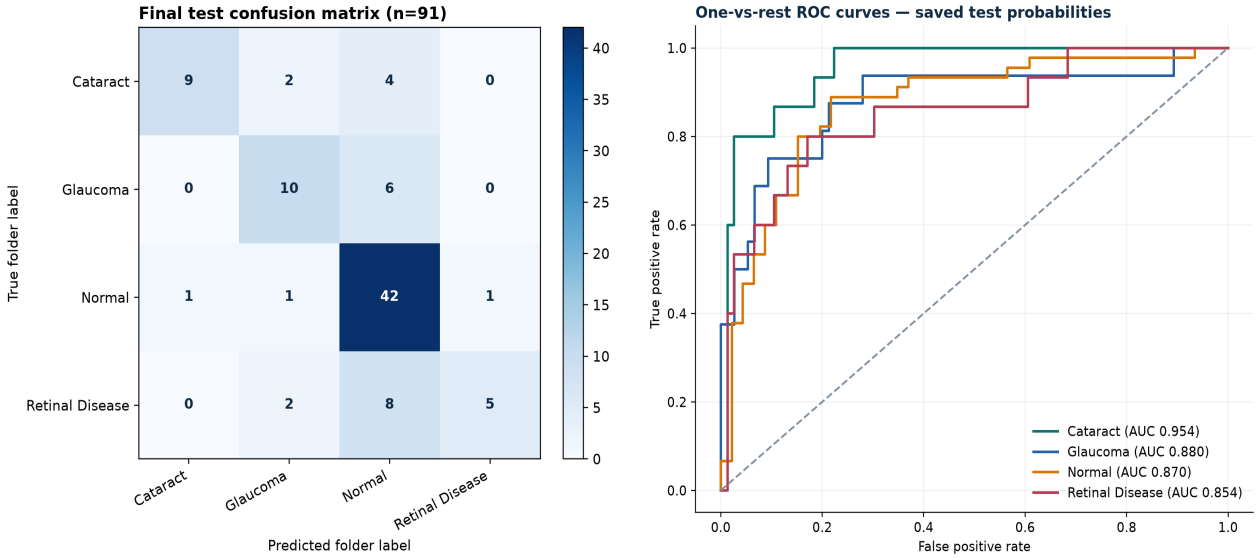


Figure 2. Confusion matrix (left) and ROC curves (right). The retinal-disease recall is 0.333; small class supports and image-level splitting limit generalization claims.

Interpretation boundary

The model predicts one of four dataset folder labels: cataract, glaucoma, normal or retinal disease. It does not grade cataract severity, fuse patient clinical records, diagnose disease, recommend treatment, or determine scheduling priority. All outputs require clinician review.

Reproducibility

Checkpoint: best macro-F1 epoch 28. Validation accuracy 68.8889%; validation macro-F1 62.7075%. Final test contains 91 images. Figures and metric tables are stored under reports/model_evaluation_v3.